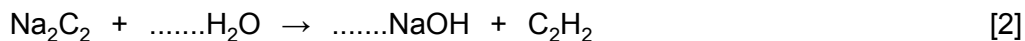


- 1 Sodium carbide, Na_2C_2 , reacts with water to form ethyne, C_2H_2 .

(a) Complete the symbol equation for this reaction.



(b) Calculate the relative formula mass of sodium carbide.

[1]

[Total: 3]

- 2 Ammonia is a base and reacts with sulfuric acid to form the salt, ammonium sulfate.

(a) What is meant by the term *base*?

..... [1]

(b) Name the industrial process used to manufacture sulfuric acid.

..... [1]

(c) Write a chemical equation for the reaction between ammonia and sulfuric acid.

..... [2]

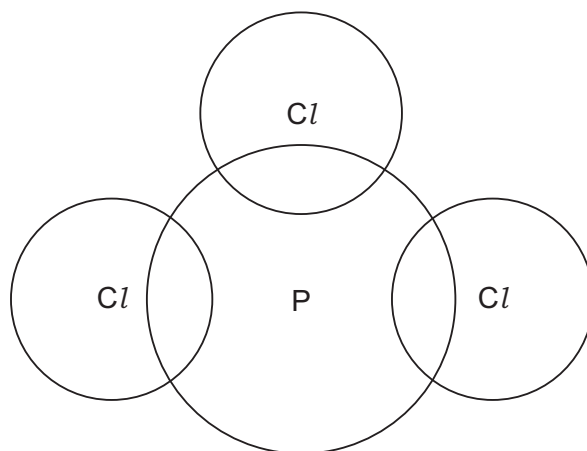
[Total: 4]

- 3 Phosphorus reacts with chlorine gas to produce phosphorus(III) chloride, PCl_3 .

(a) Write a chemical equation for the reaction between phosphorus and chlorine to produce phosphorus(III) chloride, PCl_3 .

..... [2]

- (b) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of phosphorus(III) chloride, PCl_3 . Show outer shell electrons only.



[2]

[Total: 4]

- 4 Aqueous sodium hydrogen sulfate, $\text{NaHSO}_4(\text{aq})$, contains the ions $\text{Na}^+(\text{aq})$, $\text{H}^+(\text{aq})$ and $\text{SO}_4^{2-}(\text{aq})$.

A test can be done to show the presence of $\text{SO}_4^{2-}(\text{aq})$ by adding acidified aqueous barium chloride or acidified aqueous barium nitrate.

- (a) State the observation that would show that SO_4^{2-} is present.

..... [1]

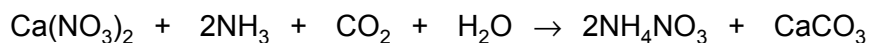
- (b) Write an ionic equation for the reaction that occurs if SO_4^{2-} is present. Include state symbols.

..... [2]

[Total: 3]

- 5 Nitrates such as ammonium nitrate are used as fertilisers.

The final stage in the production of ammonium nitrate is shown in the equation.



Calculate the maximum mass of ammonium nitrate that can be produced from 820 g of calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, using the following steps.

The relative formula mass, M_r , of calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, = 164.

(a) Calculate the number of moles of $\text{Ca}(\text{NO}_3)_2$ in 820 g.

..... mol [1]

(b) Deduce the number of moles of NH_4NO_3 produced.

..... mol [1]

(c) Calculate the M_r of NH_4NO_3 .

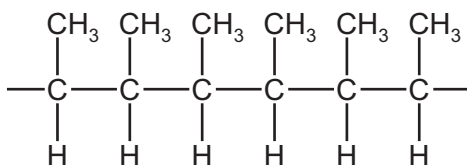
M_r of NH_4NO_3 = [1]

(d) Calculate the maximum mass of ammonium nitrate produced.

..... g [1]

[Total: 4]

6 Part of an addition polymer is shown.



- (a) How many monomer units are needed to make the part of the addition polymer shown?

..... [1]

- (b) Draw the structure of the monomer that is used to make this addition polymer. Show all of the atoms and all of the bonds.

Name the monomer.

name [2]

- (c) State the empirical formula of:

the monomer

the polymer [2]

[Total: 5]

- 7 The gases Ar, CO₂, N₂ and O₂ are in clean, dry air.

CO, NO, NO₂ and SO₂ are gases commonly found in polluted air.

- (a) What percentage of clean, dry air is N₂?

Give your answer to the nearest whole number.

..... % [1]

- (b) Name the process used to separate O₂ from clean, dry air.

..... [2]

- (c) State **one** major adverse effect of the pollutant SO₂.

..... [1]

- (d) NO and NO₂ are produced in car engines.

Describe how oxides of nitrogen form in a car engine.

.....

 [2]

- (e) Many cars have catalytic converters in their exhaust systems. In a catalytic converter, most of the CO and NO formed in a car engine is changed into less harmful products.

Identify these products and state the metal catalyst used.

products
 catalyst [3]

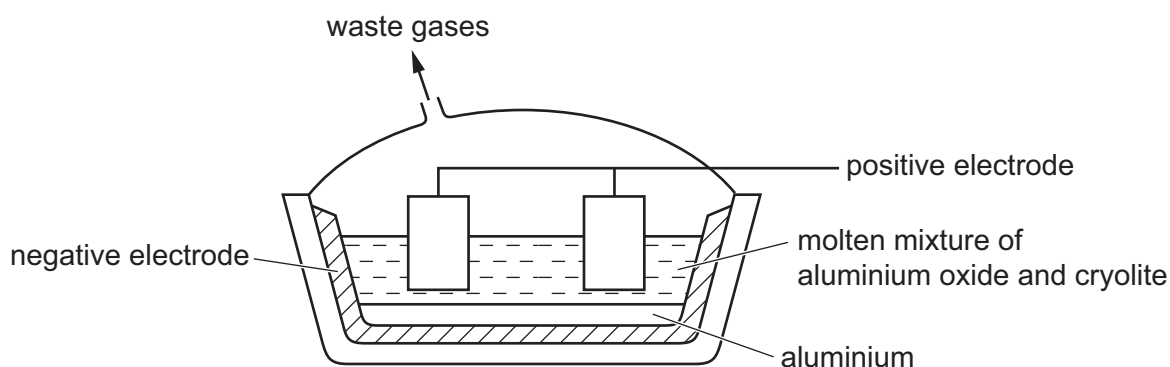
- (f) CO is formed from the incomplete combustion of fossil fuels such as methane.

Write a chemical equation to show the incomplete combustion of methane.

..... [2]

[Total: 11]

- 8 Aluminium is produced by the electrolysis of aluminium oxide dissolved in molten cryolite.



- (a) Give **two** reasons why the electrolysis is done using a molten mixture of aluminium oxide and cryolite instead of molten aluminium oxide only.

1
 2 [2]

- (b) Write ionic half-equations for the reactions occurring at the electrodes.

positive electrode
 negative electrode [2]

- (c) The anodes are made of carbon and have to be replaced regularly.

Explain why the carbon anodes have to be replaced regularly.

.....

.....

[2]

[Total: 6]

- 9 The table shows the percentage by mass of the elements in the oceans and in the biosphere. The biosphere is all living organisms.

element	percentage by mass in the oceans	percentage by mass in the biosphere
calcium	0.05	0.40
carbon	0.01	39.00
chlorine	1.80	0.05
hydrogen	11.00	6.60
magnesium	0.12	0.10
oxygen	85.80	53.00
silicon	0.00	0.10
sodium	1.15	0.05
other elements	0.07	
total	100.00	100.00

Answer these questions using only the information in the table.

- (a) Deduce the percentage by mass of other elements present in the biosphere.

..... % [1]

- (b) Which metallic element is present in the oceans in the greatest percentage by mass?

..... [1]

- (c) Give **two** major differences in the percentage by mass of the elements in the oceans and in the biosphere.

1

.....

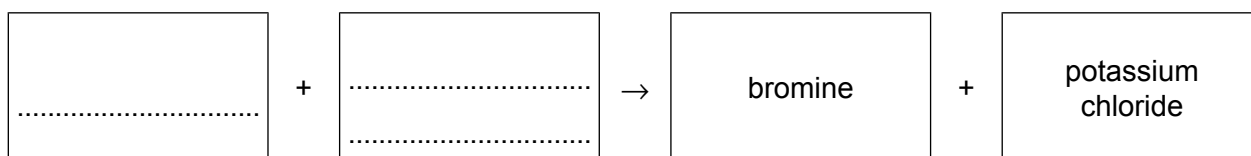
2

..... [2]

[Total: 4]

- 10** This question is about bromine and compounds of bromine.

- (a) Complete the word equation to show the halogen and halide compound which react to form the products bromine and potassium chloride.



[2]

- (b) Explain, in terms of the reactivity of the halogens, why aqueous bromine will **not** react with aqueous potassium chloride.

.....

..... [1]

[Total: 3]

- 11** The table shows the percentage by mass of the elements in the Earth's crust and in the oceans.

element	percentage by mass in the Earth's crust	percentage by mass in the oceans
aluminium	8.20	0.00
calcium	3.60	0.05
chlorine	0.05	1.80
hydrogen	0.22	11.00
iron	5.00	0.00
oxygen	46.60	85.80
silicon	29.50	0.00
sodium	2.80	1.15

other elements	4.03	
total	100.00	100.00

Answer these questions using only the information in the table.

- (a) Deduce the percentage by mass of other elements present in the oceans.

..... % [1]

- (b) Which element is present in the Earth's crust in the greatest percentage by mass?

..... [1]

- (c) Give **two** major differences in the percentage by mass of the elements in the Earth's crust and in the oceans.

1

.....

2

..... [2]

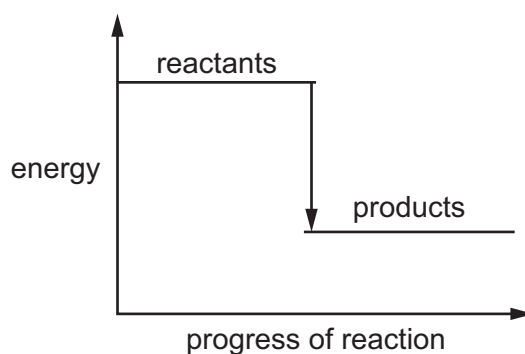
[Total: 4]

12 Bromine reacts with hydrogen sulfide, H_2S .

- (a) Complete the chemical equation for this reaction.

..... + H_2S →HBr + S [2]

(b) The energy level diagram for this reaction is shown.



Explain how this diagram shows that the reaction is exothermic.

.....
 [1]

[Total: 3]

13 Complete the table to show the number of electrons, protons and neutrons in the oxygen atom and cobalt ion shown.

	number of electrons	number of neutrons	number of protons
$^{17}_{8}\text{O}$			
$^{59}_{27}\text{Co}^{2+}$			27

[4]

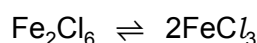
[Total: 4]

14 Iron reacts with chlorine to form iron(III) chloride, Fe_2Cl_6 .

(a) Balance the chemical equation for this reaction.



(b) At 400 °C Fe₂Cl₆ molecules decompose.



What is meant by the symbol $\rightleftharpoons$?

..... [1]

[Total: 3]

- 15** Iron can be converted into steel in a basic oxygen converter.

Oxygen is blown into the impure molten iron to remove some of the impurities.

- (a)** Oxygen reacts with the carbon in the impure iron to form carbon dioxide.

Write a chemical equation for this reaction.

..... [2]

- (b)** Basic oxides in the lining of the converter react with impurities such as sulfur dioxide to form slag.

What type of oxide is sulfur dioxide?

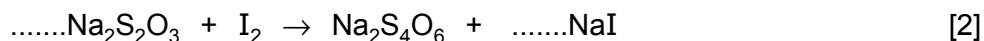
Give a reason for your answer.

..... [2]

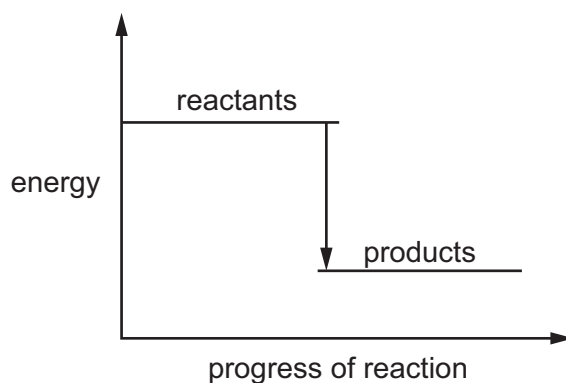
[Total: 4]

- 16** Iodine reacts with aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$.

- (a)** Balance the chemical equation for this reaction.



- (b)** The energy level diagram for this reaction is shown.



Explain how this diagram shows that the reaction is exothermic.

..... [1]

[Total: 3]

- 17 An organic compound made from sodium azide has the composition by mass: 49.5% carbon, 7.2% hydrogen and 43.3% nitrogen.

Calculate the empirical formula of the organic compound.

[3]

[Total: 3]

- 18 Silver chloride, AgCl , is insoluble. It can be made by a precipitation reaction between aqueous barium chloride and a suitable aqueous silver salt.

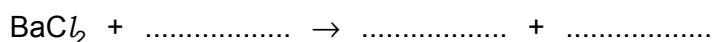
(a) What is meant by the term *precipitate*?

.....

..... [2]

- (b) Complete the chemical equation to show the formation of insoluble silver chloride from aqueous barium chloride and a suitable silver salt.

Name the silver salt you have chosen for the preparation of silver chloride



name of suitable silver salt..... [3]

[Total: 5]

- 19 Hydrochloric acid produces salts called chlorides.

Magnesium carbonate reacts with hydrochloric acid to produce magnesium chloride.



A student used 50.00 cm^3 of 2.00 mol/dm^3 hydrochloric acid in an experiment to produce magnesium chloride.

Calculate the mass, in g, of magnesium carbonate needed to react exactly with 50.00 cm^3 of 2.00 mol/dm^3 hydrochloric acid using the following steps.

- (a) Calculate the number of moles of HCl present in 50.00 cm^3 of $2.00 \text{ mol/dm}^3 \text{ HCl}$.

..... mol [1]

- (b) Determine the number of moles of MgCO_3 which would react with 50.00 cm^3 of $2.00 \text{ mol/dm}^3 \text{ HCl}$.

..... mol [1]

- (c) Calculate the relative formula mass, M_r , of MgCO_3 .

M_r of $\text{MgCO}_3 =$ [1]

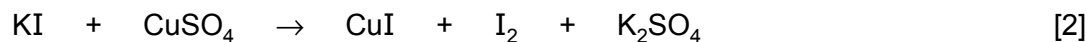
- (d) Calculate the mass of MgCO_3 needed to react exactly with 50.00 cm^3 of $2.00 \text{ mol/dm}^3 \text{ HCl}$.

mass = g [1]

[Total: 4]

- 20** Aqueous potassium iodide reacts with aqueous copper(II) sulfate to produce iodine.

- (a) Balance the chemical equation for this reaction.



- (b) Deduce the charge on the copper ion in CuI .

..... [1]

- (c) In terms of electron transfer, explain why copper is reduced in this reaction.

..... [1]

(d) Identify the reducing agent.

..... [1]

[Total: 5]

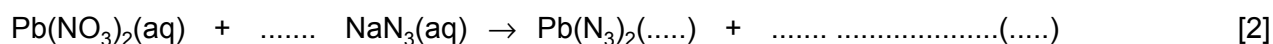
- 21** Lead(II) azide is insoluble in water. Solid lead(II) azide can be made in a precipitation reaction between aqueous lead(II) nitrate and aqueous sodium azide.

Lead(II) azide has the formula $\text{Pb}(\text{N}_3)_2$.

(a) Deduce the formula of the azide ion.

..... [1]

- (b) Complete the chemical equation for the reaction between aqueous lead(II) nitrate and aqueous sodium azide to form solid lead(II) azide and aqueous sodium nitrate. Include state symbols.



- (c) Describe how you could obtain a sample of lead(II) azide that is **not** contaminated with any soluble salts from the reaction mixture.

.....

 [2]

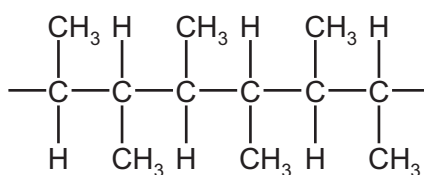
[Total: 5]

- 22** Alkenes can form polymers.

(a) What type of polymerisation occurs when alkenes form polymers?

..... [1]

(b) Part of the structure of a polymer is shown.

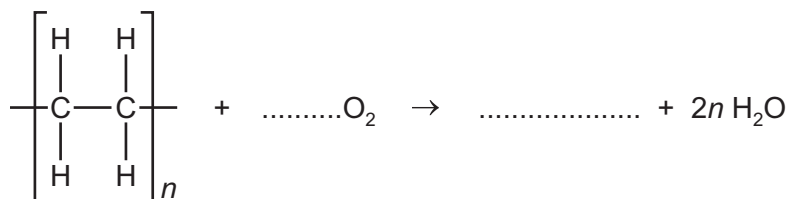


Draw the structure of the alkene from which this polymer can be made. Show all of the atoms and all of the bonds.

[1]

(c) Polymers can undergo incomplete combustion to form carbon monoxide.

Complete the chemical equation for the incomplete combustion of poly(ethene). The only carbon-containing product is carbon monoxide.



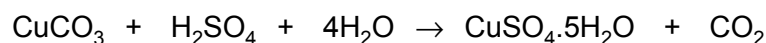
[2]

[Total: 4]

23 Copper(II) sulfate crystals, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, are hydrated.

Copper(II) sulfate crystals are made by reacting copper(II) carbonate with dilute sulfuric acid.

The equation for the overall process is shown.



step 1 Powdered solid copper(II) carbonate is added to 50.0 cm^3 of 0.05 mol/dm^3 sulfuric acid until the copper(II) carbonate is in excess.

step 2 The excess of copper(II) carbonate is separated from the aqueous copper(II) sulfate.

step 3 The aqueous copper(II) sulfate is heated until the solution is saturated.

step 4 The solution is allowed to cool and crystallise.

step 5 The crystals are removed and dried.

Calculate the maximum mass of the copper(II) sulfate crystals, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, that can form using the following steps.

- (a) Calculate the number of moles of H_2SO_4 in 50.0 cm^3 of $0.05 \text{ mol/dm}^3 \text{ H}_2\text{SO}_4$.

..... mol [1]

- (b) Determine the number of moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ that can form.

..... mol [1]

- (c) The M_r of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 250.

Calculate the maximum mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ that can form.

..... g [1]

[Total: 3]

24

Magnesium exists as three isotopes, $^{24}_{12}\text{Mg}$, $^{25}_{12}\text{Mg}$ and $^{26}_{12}\text{Mg}$.

All isotopes of magnesium react with dilute hydrochloric acid to make hydrogen and a salt.

(a) Why do all isotopes of magnesium react in the same way?

.....
.....
..... [2]

(b) Write a chemical equation for the reaction between magnesium and dilute hydrochloric acid.

..... [2]

(c) Describe a test for hydrogen.

test
result [2]

[Total: 6]

25 When chlorine reacts with aqueous potassium bromide a displacement reaction occurs.

(a) Describe the colour change of the solution.

from to [2]

(b) Write a chemical equation for this reaction.

..... [2]

[Total: 4]

26 Phosphine, PH_3 , has a similar chemical structure to ammonia, NH_3 .

Ammonia acts as a base when it reacts with sulfuric acid.

(a) What is meant by the term *base*?

..... [1]

(b) Write a chemical equation for the reaction between ammonia and sulfuric acid.

..... [2]

[Total: 3]

27 Zinc and copper are elements next to each other in the Periodic Table.

Zinc is obtained from zinc blende in a two-step process.

- In **step 1**, zinc blende is converted into zinc oxide.
- In **step 2**, zinc oxide is converted into zinc in a blast furnace.

Outline how each of these steps are done.

In your answer:

- give **one** chemical equation for each step
- describe how zinc is removed from the blast furnace in **step 2**.

step 1

.....

chemical equation

step 2.....

.....

chemical equation.....

removal of zinc in **step 2**

.....

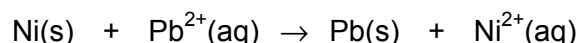
[5]

[Total: 5]

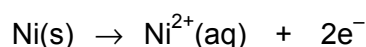
28 Displacement reactions occur between metals and metal ions.

Displacement reactions can be used to determine the order of reactivity of metals such as lead (Pb), nickel (Ni), and silver (Ag).

The ionic equation for a displacement reaction is shown.



The ionic half-equations for this reaction are shown.



The ionic half-equations show that electrons are donated by nickel atoms and accepted by

lead ions.

(a)

(i) Identify the reducing agent in the displacement reaction. Give a reason for your answer.

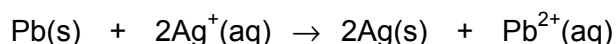
reducing agent

reason [2]

(ii) What is the general term given to the type of reaction in which electrons are transferred from one species to another?

..... [1]

(b) The ionic equation for another displacement reaction is shown.



Write the **two** ionic half-equations for this reaction.

1

2 [2]

(c) Use the information in (a) and (b) to put the three metals lead, nickel and silver in order of reactivity.

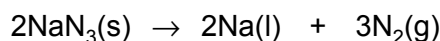
	most reactive
	↑
	least reactive

[1]

[Total: 6]

- 29** Some car airbags contain sodium azide.
When a car airbag is used the sodium azide, NaN_3 , decomposes.
The products are nitrogen and sodium.

The equation for the decomposition of sodium azide is shown.



Calculate the mass, in g, of sodium azide needed to produce 144 dm^3 of nitrogen using the following steps.

- (a) Calculate the number of moles in 144 dm^3 of N_2 measured at room temperature and pressure.

moles of N_2 =mol [1]

- (b) Determine the number of moles of NaN_3 needed to produce this number of moles of N_2 .

moles of NaN_3 =mol [1]

- (c) Calculate the relative formula mass, M_r , of NaN_3 .

M_r = [1]

- (d) Calculate the mass of NaN_3 needed to produce 144 dm^3 of N_2 .

.....g [1]

[Total: 4]

- 30** Phosphorus, P_4 , reacts with air to produce phosphorus(V) oxide, P_4O_{10} .

- (a) Write a chemical equation for this reaction.

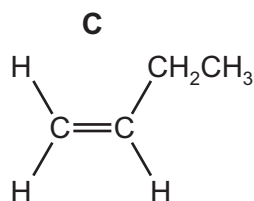
..... [2]

- (b) What type of chemical reaction is this?

..... [1]

[Total: 3]

31 Alkene **C** can be converted into a polymer.



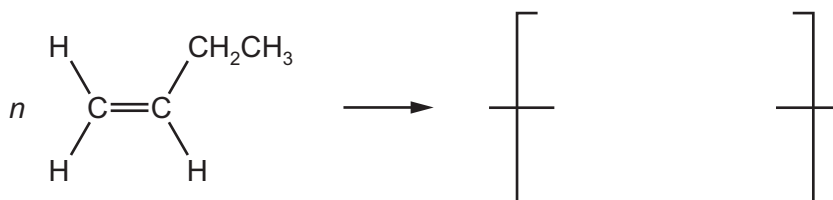
(a) What type of polymerisation occurs?

..... [1]

(b) Suggest the name of the polymer formed.

..... [1]

(c) Complete the chemical equation to show this polymerisation.



[3]

(d) State the empirical formula of the polymer formed.

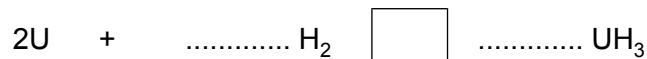
..... [1]

[Total: 6]

32 Uranium reacts with hydrogen to form uranium hydride, UH_3 .
The reaction is reversible.

Complete the chemical equation for this reaction by:

- balancing the equation
- drawing the symbol for a reversible reaction in the box.



[3]

[Total: 3]

33 This question is about sulfur, sulfur compounds and the water from a sulfur spring. A sulfur spring is a natural source of water containing sulfur.

The table shows the mass of ions present in a 1000 cm^3 sample of water from a sulfur spring.

ion present	formula of ion	mass present in the 1000 cm ³ sample / mg
	Br ⁻	4
calcium	Ca ²⁺	44
chloride	Cl ⁻	14
fluoride	F ⁻	6
iron(III)	Fe ³⁺	2
magnesium	Mg ²⁺	10
	K ⁺	8
sodium	Na ⁺	88
sulfate	SO ₄ ²⁻	92

Answer these questions using only information from the table.

- (a) Which negative ion is present in the lowest concentration?

..... [1]

- (b) Give the name of the compound formed from only K⁺ and Br⁻ ions.

..... [1]

- (c) Calculate the mass of calcium ions present in a 250 cm³ sample of this water.

mass of calcium ions = mg [1]

- (d) Complete the equation to show the formation of a fluoride ion from a fluorine atom.

F + → F⁻ [1]

[Total: 4]

- 34 Tetrafluoromethane and lead(II) fluoride are fluorides of Group IV elements. Some properties of tetrafluoromethane and lead(II) fluoride are shown in the table.

property	tetrafluoromethane	lead(II) fluoride
formula	CF ₄	
melting point / °C	-184	855
boiling point / °C	-127	1290

conduction of electricity when solid	non-conductor	non-conductor
conduction of electricity when molten	non-conductor	good conductor

(a) What is the formula of lead(II) fluoride?

..... [1]

(b) What type of bonding is present between the atoms in tetrafluoromethane?

..... [1]

(c) What type of structure does solid lead(II) fluoride have?

..... [1]

(d) Explain, in terms of attractive forces between particles, why lead(II) fluoride has a much higher melting point than tetrafluoromethane.

In your answer refer to the types of attractive forces between particles and their relative strengths.

.....

 [3]

[Total: 6]

35 Rust contains iron(III) oxide.

Phosphoric acid, H_3PO_4 , can be used to remove rust from an iron object and prevent further rusting.

(a) Write a chemical equation for the reaction between iron(III) oxide and phosphoric acid to form iron(III) phosphate and water.

..... [2]

(b) Iron(III) phosphate is an insoluble salt.

Suggest how the formation of iron(III) phosphate prevents further rusting.

.....
 [1]

[Total: 3]

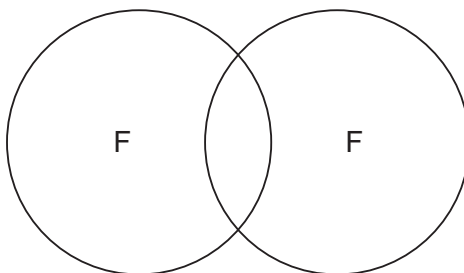
36 Fluorine is a Group VII element. Fluorine forms compounds with metals and non-metals.

- (a) Predict the physical state of fluorine at room temperature and pressure.

..... [1]

- (b) Fluorine exists as diatomic molecules.

Complete the dot-and-cross diagram to show the electron arrangement in a molecule of fluorine.
Show outer shell electrons only.



[2]

- (c) Write a chemical equation for the reaction between sodium and fluorine.

..... [2]

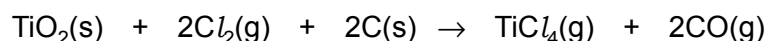
- (d) Explain why chlorine does **not** react with aqueous sodium fluoride.

..... [1]
.....

[Total: 6]

- 37** Titanium is extracted from an ore called rutile. Rutile is an impure form of titanium(IV) oxide, TiO_2 .

Calculate the volume of chlorine gas, $\text{Cl}_2(\text{g})$, at room temperature and pressure, that reacts completely with 400 g of $\text{TiO}_2(\text{s})$ using the following steps.



- (a) Calculate the relative formula mass, M_r , of TiO_2 .

M_r of TiO_2 = [1]

(b) Calculate the number of moles in 400 g of TiO_2 .

..... mol [1]

(c) Determine the number of moles of Cl_2 that react with 400 g of TiO_2 .

moles of Cl_2 = mol [1]

(d) Calculate the volume of Cl_2 that reacts with 400 g of TiO_2 .

volume of Cl_2 = dm^3 [1]

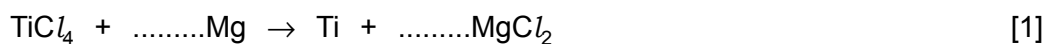
[Total: 4]

38 Titanium is extracted from an ore called rutile. Rutile is an impure form of titanium(IV) oxide, TiO_2 .

Rutile is mixed with coke and heated in a furnace through which chlorine gas is passed. The product is gaseous titanium(IV) chloride, TiCl_4 .

Titanium(IV) chloride is heated with an excess of magnesium, in an atmosphere of argon.

(a) Balance the chemical equation for the reaction.



(b) Titanium(IV) chloride can be reacted with sodium instead of magnesium.

The reaction between titanium(IV) chloride and sodium is similar to the reaction between titanium(IV) chloride and magnesium.

Write a chemical equation for the reaction between titanium(IV) chloride and sodium.

..... [1]

- (c) Suggest why the reaction between titanium(IV) chloride and magnesium is done in an atmosphere of argon and **not** in air.

.....
 [1]

[Total: 3]

- 39 Tetrafluoroethene is an unsaturated compound with the formula C_2F_4 .

Tetrafluoroethene is the monomer used to make the polymer poly(tetrafluoroethene).

- (a) What is meant by the term *unsaturated*?

.....
 [1]

- (b) Describe a test to show that tetrafluoroethene is unsaturated.

test
 observations [2]

- (c) Draw the structure of a molecule of tetrafluoroethene. Show all of the atoms and all of the bonds.

[1]

- (d) Tetrafluoroethene can be polymerised to form poly(tetrafluoroethene).

Draw **one** repeat unit of poly(tetrafluoroethene). Show all of the atoms and all of the bonds.

[2]

- (e) Deduce the empirical formula of:

tetrafluoroethene
 poly(tetrafluoroethene) [2]

[Total: 8]

- 40 How would you show that an aqueous solution of ethanoic acid, CH_3COOH , is an acid **without** using an indicator or measuring the pH?

State the reagent you would use and give the expected observations.

Write a chemical equation for the reaction that you describe.

reagent

expected observations

.....

chemical equation [3]

[Total: 3]